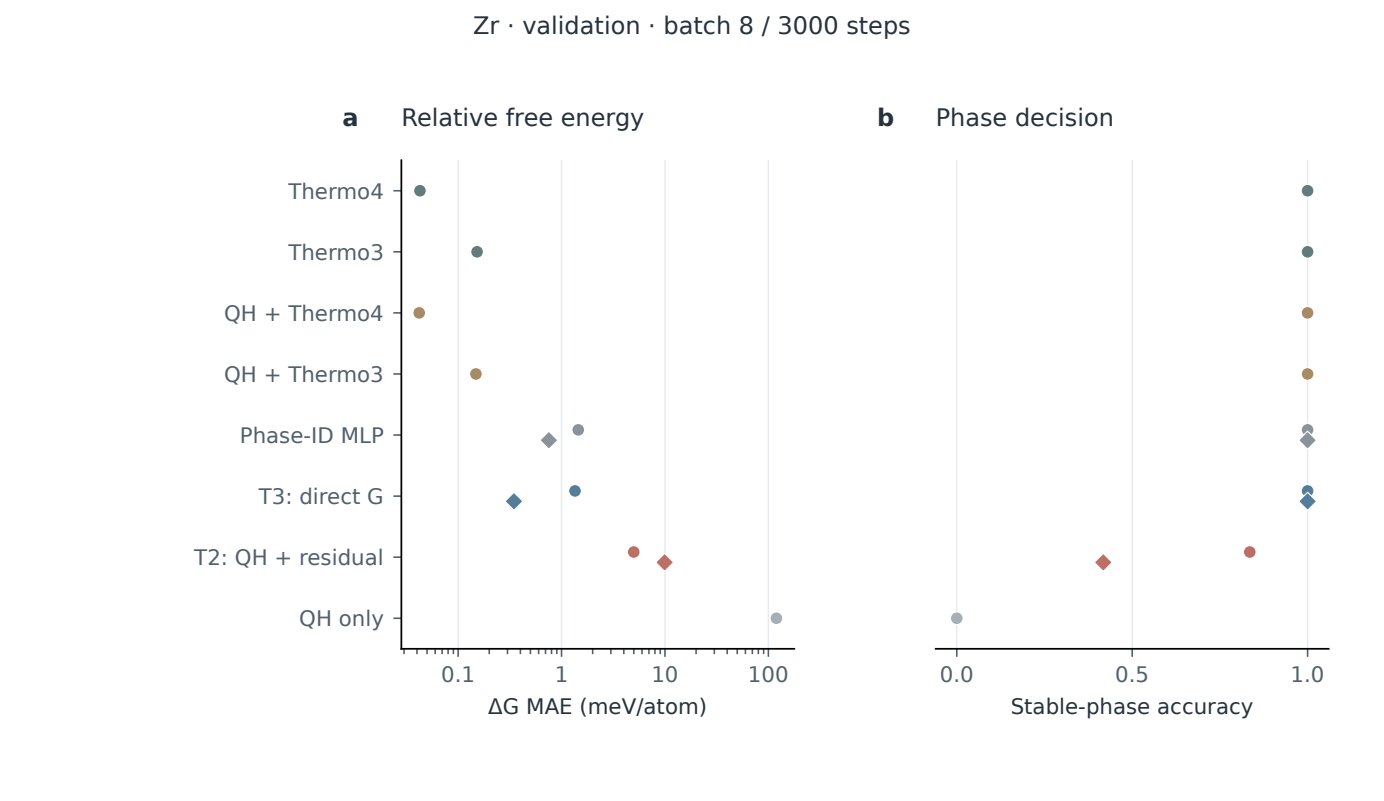
Zr · validation · batch 8 / 3000 steps
a
Relative free energy
Thermo4
Thermo3
QH + Thermo4
QH + Thermo3
Phase-ID MLP
T3: direct G
T2: QH + residual
QH only
0.1
1
10
100
ΔG MAE (meV/atom)
b
Phase decision
0.0
0.5
1.0
Stable-phase accuracy